Class XII Maths Sample Paper – Set 4

Time Allowed: 30 Minutes

General Instructions:

1. This Question paper contains - five sections A, Each section is compulsory.

However, there are internal choices in some questions.

2. Section A has 18 MCQ's and 02 Assertion-Reason based questions of 1 mark each.

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Q. No.	Questions	Marks
	Section A	
1.	The principal value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ is: (A) $-\frac{\pi}{3}$	1

	(B) $-\frac{\pi}{4}$ (C) $\frac{5\pi}{3}$ (D) $\frac{\pi}{4}$	
2.	If $f(x) = \frac{x-1}{x-2}$ (with $x \neq 2$) is an invertible function, then $f^{-1}(x)$ equals: (A) $\frac{1+2x}{x-1}$ (B) $\frac{2x-1}{x-1}$ (C) $\frac{x-2}{x-1}$ (D) $\frac{2x-1}{1-x}$	1
3.	If A is a 3×3 matrix with $\det(A)=3$, then $\det(2A)$ is: (A) 6 (B) 12 (C) 24 (D) 27	1

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4.	For what value of k is the function $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ k, & x = 2 \end{cases}$ continuous at $x = 2$? (A) $k = -4$ (B) $k = 0$ (C) $k = 2$ (D) $k = 4$	1
5.	Which of the following functions is not differentiable at $x = 0$? (A) $f(x) = e^x$ (B) $f(x) = \sin x$ (C) $f(x) = \lvert x \rvert$ (D) $f(x) = x^2$	1
6.	If $f'(x) > 0$ for all x in the interval (a, b), then $f(x)$ is: (A) increasing on (a, b) (B) decreasing on (a, b) (C) constant on (a, b) (D) periodic on (a, b)	1
7.	The line $y = 4x - 5$ is a tangent to the curve $y = x^2$ at which point?	1

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	(A) $(2,;4)$ (B) $(1,;1)$ (C) $(2,;-1)$ (D) $(3,;4)$	
8.	$\int xe^x, dx$ equals: (A) $(x-1)e^x + C$ (B) $(x+1)e^x + C$ (C) $xe^x + C$ (D) $e^x + C$	1
9.	$\int_{-5}^{5} x^3, dx$ is equal to: (A) 0 (B) 125/4 (C) 125/2 (D) 250	1
10.	The order and degree (respectively) of the differential equation $\left(\frac{d^2y}{dx^2}\right)^4 + \left(\frac{dy}{dx}\right)^2 = 5$	1

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	are: (A) 2, 4 (B) 4, 2 (C) 2, 2 (D) 4, 4	
11.	If $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cup B) = 0.7$, then $P(A \mid B)$ is: (A) 0.2 (B) 0.4 (C) 0.5 (D) 0.7	1
12.	A fair coin is tossed 4 times. The probability of getting exactly 2 heads is: (A) $\frac{3}{8}$ (B) $\frac{1}{2}$ (C) $\frac{6}{16}$ (D) $\frac{4}{16}$	1
13.	For what value of λ are the vectors $\mathbf{a} = \langle \lambda,, 2,, 1 \rangle$ and $\mathbf{b} = \langle 3,, \lambda,, -1 \rangle$ perpendicular?	1

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	(A) $\lambda = -5$ (B) $\lambda = \frac{1}{5}$ (C) $\lambda = \frac{1}{3}$ (D) $\lambda = \frac{1}{5}$	
14.	The equation of the plane passing through the point $(1, 2, 3)$ with normal vector $\mathbf{n} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ is: (A) $2x - y + 2z - 3 = 0$ (B) $2(x-1) - (y-2) + 2(z-3) = 0$ (C) $2x - y + 2z + 3 = 0$ (D) $2x + 2y + 2z - 3 = 0$	1
15.	In a linear programming problem, the optimal value (maximum or minimum) of the objective function occurs: (A) at the center of the feasible region (B) at a corner point (vertex) of the feasible region (C) at infinitely many points in the feasible region (D) only along the boundary line of the feasible region	1

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16.	The distance of the point $(3,,1,,-1)$ from the plane $x+2y-2z+1=0$ is: (A) $\frac{8}{3}$ (B) 8 (C) $\frac{4}{3}$ (D) $\frac{8}{\sqrt{6}}$	1
17.	If two events A and B are independent, then which of the following is always true? (A) $P(A\cup B)=P(A)+P(B)$ (B) $P(A\cap B)=P(A),P(B)$ (C) $P(A\mid B)=P(A\cap B)$ (D) $P(B\mid A)=P(A),P(B)$	1
18.	If μ and σ^2 are the mean and variance of a binomial distribution $B(n,p)$, then: (A) $\mu=np,;\sigma^2=np(1-p)$ (B) $\mu=np,;\sigma^2=n^2p^2$ (C) $\mu=p,;\sigma^2=p(1-p)$	1

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	(D) $\mu = np, ; \sigma^2 = \sqrt{np(1-p)}$	
	(Assertion-Reason): Questions 19 and 20 are Assertion-Reason type. Each question contains a statement of Assertion (A) and a statement of Reason (R). Select the one correct option out of the choices (A), (B), (C), (D) given below: (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is not the correct explanation of A. (C) A is true, but R is false. (D) A is false, but R is true.	1
19.	Assertion (A): If $f'(a) = 0$ and $f''(a) > 0$ for a function f, then f has a local minimum at $x = a$. Reason (R): $f'(a) = 0$ implies that $x = a$ is a stationary point of $f(x)$.	1
20.	Assertion (A): If two events A and B are mutually exclusive and each has non-zero probability, then A and B are independent events. Reason (R): If A and B are mutually exclusive, then $P(A \cap B) = 0$, but for independent events $P(A \cap B) = P(A)P(B)$.	1

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